

Mainak Kumar

Noida, India | damiankumar2000@gmail.com | +91-6291067398 | linkedin.com/in/mainak-kumar-9609761b1 | github.com/DamianKumar7

PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years building fault-tolerant, high-throughput distributed systems in Java and Spring Boot at Fortune-500 scale. Led an Auth0 authentication migration for 500+ enterprise customers handling 300K+ concurrent users with zero downtime, and delivered AML/KYC/PCI-DSS payment compliance systems and Kafka-driven event architectures in FinTech. Growing expertise in AI and agentic engineering (Claude API, LLM workflows) as a complementary edge to core backend and distributed-systems strength.

TECHNICAL SKILLS

Languages: Java (OOP, Multithreading, Concurrency, JVM), Python, SQL, JavaScript, HTML/CSS, TypeScript

Backend & Frameworks: Spring Boot, Spring MVC, Hibernate/JPA, Microservices, RESTful APIs, gRPC, GraphQL, OAuth 2.0, Auth0, Node.js, Nest.js

FinTech & Compliance: AML (Anti-Money Laundering), KYC, PCI DSS, IAM, Fraud & Risk Management, Secure Coding, Regulatory Compliance

Databases & Caching: PostgreSQL, MySQL, Redis, H2, Query Optimization, Database Indexing, Connection Pooling

Messaging & Distributed Systems: Apache Kafka, RabbitMQ, Amazon SQS, Event-Driven Architecture, Distributed Systems

Cloud, DevOps & CI/CD: Google Cloud Platform (GCP), Docker, Kubernetes, GitHub Actions, Ansible, Terraform, Linux/Unix

Observability & AI: Grafana, DataDog, Monitoring & Alerting, Claude API, LLM Integration, Agentic Workflows

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Ultimate Kronos Group (UKG) — Fortune 500, Noida, India

Sep 2024 – Present

Stack: Java, Spring Boot, JWT, OAuth 2.0, Redis, PostgreSQL, Kafka, Ansible, Terraform, GCP

- Designed an end-to-end authentication migration framework to Auth0 for 500+ enterprise customers, leading a team of 5+ engineers with a zero-downtime strategy — handling 300K+ concurrent users with full fault tolerance.
- Led priority migration of the WFM service from RightScale to GitHub Actions using Ansible and Terraform, saving \$30M/year across 10M+ customer workloads.
- Drove a code-signing security initiative (CrowdStrike) reducing vulnerabilities by 70%; built a Java REST API automating bulk credential rotation in Google Secret Manager and migrated legacy Java 8 → Java 17.

Software Engineer | PineLabs — FinTech / Payments, Noida, India

Apr 2023 – Aug 2024

Stack: Java, Spring Boot, Kafka, REST APIs, Microservices, PostgreSQL

- Architected Anti-Money-Laundering (AML) and Risk-Management microservices on a Kafka-driven event architecture, running real-time transaction monitoring across 100,000+ merchants.
- Spearheaded KYC onboarding automation and security hardening across the payments codebase, achieving PCI-DSS certification for the monolithic payments platform.
- Designed high-throughput, fault-tolerant payment processing pipelines with Kafka, ensuring zero-delay transaction handling under peak financial load.

Software Engineer | Thoughtworks, Noida, India

Feb 2022 – Mar 2023

Stack: Java, Spring Boot, Kafka, React.js, REST APIs, Microservices

- Led migration of a legacy codebase to a microservices architecture — authored technical design documents, conducted architecture reviews, and mentored junior engineers on distributed systems best practices.
- Built a Kafka event-driven pipeline for production reliability and horizontal scalability; delivered full-stack enhancements (React.js + Spring Boot) from requirements through production deployment.

TECHNICAL PROJECTS

RowOne Asia — Live Event Ticketing Platform

Next.js, Nest.js, GraphQL, PostgreSQL · rowone.asia/home

- Built and shipped a production ticketing SaaS platform serving 500,000+ active users; architected a GraphQL API layer reducing over-fetching by 40% vs. REST.
- Implemented TTL-based optimistic seat-locking to prevent double-booking under high-concurrency demand drops.

High-Throughput URL Shortener

Java, Spring Boot, PostgreSQL, Redis · github.com/DamianKumar7/urlshortener

- Engineered a distributed service with Base62 short codes and Redis caching (sub-5ms redirects, 80% DB-load reduction) on an indexed PostgreSQL schema with click analytics.

EDUCATION

B.Tech — Computer Science & Engineering

Meghnad Saha Institute of Technology, Kolkata · 2018 – 2022 · CGPA 9.1/10